Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security – Semester III

Term Assessment – I (September- 2025)

Subject Code: CTMSCS SIII P2 **Date:** 09.09.2025

Subject Name: IoT Security and Forensics

Time Duration: 45 minutes **Total Marks: 25**

Instructions:

***Attempt any five questions.**
*** Short Answer Questions**

2x5 = 10

Q.1 Define IoT in one sentence.?

Q.2 List two wireless protocols commonly used in IoT.?

Q.3 Name three key characteristics of IoT devices.?

Q.4 What does M2M communication stand for?

Q.5 What is edge computing in IoT context

Q.6 Name one major security challenge in IoT systems?

***Long Answer Questions**
***Attempt any three questions.**

3*5=15

Q.7 Explain the difference between physical and logical design in IoT systems.?

Q.8 Identify and explain three major challenges facing IoT implementation.?

Q.9 Describe three different IoT deployment levels and their applications.?

Q.10 Compare and contrast two IoT enabling technologies.?

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NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security, Sem - III

Term Assessment – I (September- 2025)

Subject Code: CTMSCS SIII P1 **Date:** 08.09.2025

Subject Name: Blockchain and Cryptocurrency

Time Duration: 45 minutes **Total Marks: 25**

Instructions: All questions are compulsory

Short Answer Questions

1x10 = 10

Q.1 What is one way function?

Q.2 Define hashing with an example.

Q.3 What is a hash pointer?

Q.4 Differentiate between Public and Private blockchains.

Q.5 What is the function of the mempool?

Q.6 What is a Genesis Block?

Q.7 What does anonymity mean in blockchain?

Q.8 What is longesh chain rule.

Q.9 What is the difference between a soft fork and a hard fork?

Q.10 List the component of Blockchain Newtork.

Long Answer Questions

Attempt any three questions: **5x3 =15**

Q.1 Explain ECDSA with an example.

Q.2 Explain the concept of a Merkle Root in blockchain and its role in transaction verification. A Merkle tree is constructed with 56 transaction leaves. Determine the number of sibling hashes required to prove inclusion of one transaction.

Q.3 What is double spending? Explain any two attack methods used for double spending.

Q.4 Describe the life cycle of a blockchain transaction

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security

(Semester 3)

Term Assessment -1 (September 2025)

Subject Code: CTMSCS SIII P3 **Date:** 10-09-2025

Subject Name: Cloud Security and Forensics

Time: 11:00 AM to 11:45 AM **(45 minutes)** **Total Marks:** 25

Instructions:

All questions are compulsory.

Short Answer Questions **(1 x 10 = 10)**

***Attempt all the questions.**

Q1. The concept of paying for computing resources as needed, similar to electricity, is known as _____ computing.

Q2. In cloud computing, _____ allows multiple virtual machines to run on a single physical server.

Q3. The characteristic of cloud computing that allows services to be accessed via phones, tablets, and laptops is called _____.

Q4. A startup needs to deploy applications quickly and scale them based on user demand without managing physical servers. They choose a cloud model that offers infrastructure on demand. This model is called _____.

Q5. In the _____ cloud model, the infrastructure is shared by several organizations and supports a specific community with shared concerns.

Q6. A government agency wants to host sensitive citizen data and requires full control over its infrastructure. It also has financial and technical resources to manage its own cloud environment. The most suitable deployment model for this agency is the _____ cloud.

Q7. The cloud service model where users access applications hosted by the provider via a web browser is known as _____.

Q8. The cloud computing model that provides platform independency and enables mobile and collaborative business applications is _____.

Q9. A forensic investigator is tasked with analyzing a breach in a hybrid cloud environment. To ensure compliance and traceability, the investigator uses a service that provides tools for evidence collection, chain of custody, and audit trails in cloud systems. This service is referred to as _____.

Q10. An organization wants to avoid full responsibility for cloud maintenance but still collaborate with similar institutions. The ideal deployment model for this scenario is community.

Long Answer Questions (5 x 3 = 15)

***Attempt any 3 questions.**

Q11. Explain the five essential characteristics of cloud computing as defined by NIST. How do these characteristics contribute to the efficiency and scalability of cloud services? Support your answer with an example(s).

Q12. A financial institution experiences a data breach in its hybrid cloud setup. The internal security team is unsure whether to rely on traditional forensic methods or adopt Cloud Forensics as a Service (CFaaS). As a forensic expert, critically assess the limitations (at least 5) of traditional forensics in cloud environments and explain how CFaaS can address these challenges. Provide an example of a scenario where CFaaS proved beneficial.

Q13. Explain the advantages and disadvantages of the public cloud model with examples.

Q14. An organization wants to collaborate with other institutions on a joint research project but lacks the budget for a private cloud. Which deployment model should they adopt and why? community.

private
public
hybrid
community

resource pooling
measured services
rapid elasticity
BAN
on demand

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security- Semester III

Term Assessment – I September- 2025

Subject Code: CTMSCS SIII P5 EL1 **Date:** 12.09.2025

Subject Name: Social Network Analysis

Time: 45 minutes **Total Marks: 25**

Instructions: All questions are compulsory

Short Answer Questions

(2*5=10)

Q.1. How can proxy or VPN complicate cyber crime investigations involving social networks?

Q.2. Given the following adjacency list representation of a network:

- A → B, C
- B → A, D
- C → A, D
- D → B, C

Convert this adjacency list into a graph representation and explain the connections among the nodes.

Q.3. Are social media and social network the same? Discuss.

Q.4. Differentiate ASINT and OSINT.

Q.5. Define cliques and geodesic distance in graph theory.

Long Answer Questions

Attempt any three questions **(5*3 =15)**

Q.1. Elaborate various crimes associated with social media and social network.

Q.2. Explain different centrality measures and its importance in social network analysis.

Q.3. During a cybersecurity audit, you suspect that sensitive organizational may be publicly accessible on the internet due to misconfigurations.

a. Explain how Google Dorking can be used to identify such exposed resources.
b. Provide examples of Google search operators (dorks) that could be useful in locating sensitive files or login pages.

Q.4. You are part of a digital forensics investigation team analyzing an image recovered from a suspect's device. The suspect claims the photo was taken in India, but investigators suspect it was downloaded from the internet.

a. Explain how you would use reverse image analysis techniques to verify whether the image is original or sourced online.
b. If the reverse image search shows visually similar results from different websites, how would you determine the most likely source and original context of the image?

National Forensic Sciences University
School of Cyber Security and Digital Forensics

Course Name: M. Sc. Cyber Security, III Semester

Subject Name: IoT Security and Forensics

Subject Code: CTMSCS SIII P2

Exam: Mid Semester Examination (October - 2025)

Time: 11.00 am to 12.30 pm

Date: 07.10.2025

Q1. Answer the following questions in short. (Attempt any 5) **[25 Marks]**

1) Explain the importance of connectivity in IoT.

2) What are the main characteristics of IoT?

3) Explain the IoT Levels in brief.

4. What are the major challenges in IoT?

5. List out all the physical components of IoT.

6. Define IoT Deployment Techniques.

Q2. Answer the following questions in detail. (Attempt any 2) **[16 Marks]**

1) Explain how scalability, interoperability, and security influence IoT development.

3) Explain the physical design and logical design of IoT.

4) What are the major challenges faced in IoT? Discuss in detail.

Q3. Select the Correct answer from the options (Attempt any 9) **[09 Marks]**

1. **Which of the following best describes IoT?**
A. A network of computers
B. A network of interconnected devices that communicate and share data
C. A type of cloud storage
D. A programming language

2. **In IoT, devices communicate using which of the following?**
A. Internet Protocol (IP)
B. Bluetooth
C. ZigBee
D. All of the above

3. Which of the following is not a component of physical design in IoT?

A. Sensors
B. Actuators
C. Data Analytics
D. Connectivity

4. How many levels of IoT are defined based on complexity and function?

A.Two Level
B.Six Level
C.Three Level
D.One Level

5. Logical design of IoT focuses on:

A) Physical devices
B) Abstract architecture and data flow
C) Network cables
D) Hardware connections

6. IoT enabling technologies include:

A. Cloud Computing
B. Big Data
C. Artificial Intelligence
D. All of the above

7. Which level uses cloud for storage and processing?

A) Level 2 (Device-to-Cloud)
B) Level 3 (Device-to-Gateway)
C) Level 5
D) Level 6

8. Edge computing is mainly used to:

A) Store data centrally
B) Process data near the source device
C) Replace all sensors
D) Remove the need for the internet

9. IoT evolved from which earlier concept?

A) Social Networking
B) Machine-to-Machine (M2M) Communication
C) Artificial Intelligence
D) Virtual Reality

10. Who introduce the term "Internet of Things"?

A) Tim Berners-Lee
B) Kevin Ashton
C) Mark Weiser
D) Bill Gates

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.Sc. Cyber Security

Mid Semester Examination, Semester - III - October -2025

Subject Code: CTMSCS SIII P1 **Date:**09/10/2025
Subject Name: Blockchain and Cryptocurrency
Time: 11:00 AM TO 12:30 PM **Total Marks: 50**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

Q.1.	Answer the following / Define the following (a) Differentiate between hashing and encryption. (b) What is a DAG (Directed Acyclic Graph)? (c) Define zero-knowledge proof. (d) List properties of blockchain. (e) How consortium blockchain different form private blockchain? (f) What is the difference solo mining and pool mining? (g) What is Coinbase transaction? (h) How is the difficulty level in blockchain mining decided? (i) How does Nakamoto Consensus achieve agreement in blockchain? (j) How do Bitcoin incentives motivate miners and discourage attackers?	10*2 (Marks)=20
Q.2.	Answer the following question in short (any 4) (a) Discuss advantages of blockchain over conventional distributed database. (b) Explain the role and tasks of miners in blockchain mining. (c) Explain different types of mining based on hardware? (d) Explain how Proof of Burn (PoB) works in blockchain and analyze its advantages and challenges. (e) Explain the Byzantine Generals Problem in the context of distributed systems or blockchain. How it affects consensus in decentralized networks?	4*5 (Marks)=20
Q.3.	Answer the following in detail Explain Proof of Stake (PoS) in blockchain, how it works to validate transactions and create new blocks, and compare it with Proof of Work (PoW). OR Explain what a Sybil attack is in the context of blockchain. Describe the different types of Sybil attacks, how they can compromise the security and integrity of a blockchain network.	1*10 (Marks)=10

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

School of Cyber Security and Digital Forensics

M.Sc. Cyber Security-Semester-III

Mid Semester Examination – October 2025

Subject Code: CTMSCS SIII P5 EL 1 **Date:10-10-2025**
Subject Name: Social Network Analysis
Time: 1 hour 30 min **Total Marks: 50**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Figures to the right indicate full marks.**

Q.I. Answer the following (Any five) **5*5 =25 Marks**

1. Differentiate between surface web, Dark web and Deep web.
2. Explain Google Dorking with an example.
3. Explain different types social media crimes.
4. Explain the role of metadata in online intelligence gathering. Name some tools that can be used to extract metadata.
5. Explain IT Act, 2000 and its various provisions related to social media crimes.
6. Discuss the psychological mechanisms and social contexts that enable the rapid spread of misinformation and fake news.

Q. II. Answer the following questions in detail (any 2) **8*2= 16 Marks**

1. What do you understand by fake news? Explain the process of fake news detection.
2. Explain the different types of centrality measures and its uses in social network analysis.
3. Compare and contrast Digital Personal Data Protection (DPDP) Act, 2023 and the General Data Protection Regulation (GDPR) of EU.

Q.III. Answer the following: **9 Marks**

1. Give some examples of metadata in online multimedia. (1)
2. What is the difference between in-degree and out-degree in a social network? (1)
3. What is sentiment analysis? (1)
4. Differentiate between OSINT and ASINT. (2)
5. Give an example of adjacency matrix representation of a graph. (2)
6. Explain reverse image search briefly. (2)

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security

(Semester 3)

Mid-Semester Examination

(October 2025)

Subject Code: CTMSCS SIII P3 **Date:** 08-10-2025

Subject Name: Cloud Security and Forensics

Time: 11:00 AM to 12:30 PM **(90 minutes)** **Total Marks:** 50

Instructions:

- All questions are compulsory.
- For fill in the blanks and objective questions only write the answer(s).

Fill in the Blanks (1 x 10 = 10)

***Attempt all the questions.**

Q1. _____ is the cloud characteristic that allows services to be accessed from anywhere using any device.

Q2. _____ is a software layer that enables virtualization in cloud environments.

Q3. _____ is the process of collecting and analyzing digital evidence from cloud environments.

Q4. _____ ensures that digital evidence remains unchanged and untampered.

Q5. During a cybercrime investigation, the forensic team must maintain a proper _____ to ensure evidence validity.

Q6. When a cloud provider offers APIs for service access and billing based on usage, it is called _____.

Q7. In cloud computing, _____ refers to the ability to scale resources up or down as needed.

Q8. A company uses Docker and Kubernetes for deployment. This is part of the _____ revolution.

Q9. A forensic investigator uses Magnet AXIOM to analyze cloud data. This tool supports _____ analysis.

Q10. A deleted file is recovered from a cloud image. This process is part of _____.

Objective Questions

(1 x 10 = 10)

*Attempt all the questions.

Q11. Which of the following is **NOT** a cloud characteristic?

a) **Resource Pooling**
b) **Rapid Elasticity**
c) **Data Mining**
d) **Measured Service**

Q12. Which model allows users to deploy applications without managing infrastructure?

a) **IaaS**
b) **SaaS**
c) **PaaS**
d) **DaaS**

Q13. Which of the following is a type of hypervisor?

a) **Type-1**
b) **Type-2**
c) **Both**
d) **None**

Q14. A company wants to run legacy applications in the cloud. Which model should they choose?

a) **SaaS**
b) **PaaS**
c) **IaaS**
d) **XaaS**

Q15. A business needs high security and full control over its cloud infrastructure. Which model is ideal?

a) **Public Cloud**
b) **Private Cloud**
c) **Hybrid Cloud**
d) **Community Cloud**

Q16. A company uses AWS and Azure simultaneously. This is called:

a) **Hybrid Cloud**
b) **Multi-Cloud**
c) **Private Cloud**
d) **Public Cloud**

Q17. A deleted file is not found in the cloud image. What is the limitation?

a) **Lack of storage**
b) **No backup**
c) **Deleted from recycle bin**
d) **Encryption**

Q18. A company wants to avoid downtime during peak traffic. They should use:

a) **Cloud Bursting**
b) **SaaS**
c) **VPN**
d) **Hypervisor**

Q19. A forensic report must include metadata and timestamps. This is part of:

a) **Identification**
b) **Reporting**
c) **Preservation**
d) **Collection**

Q20. What does the term "Chain of Custody" refer to in cloud forensics?

a) **A method of encrypting cloud data**
b) **A list of cloud service providers**
c) **A documented trail of evidence handling**
d) **A type of cloud deployment model**

Short Answer Questions

(3 x 2 = 6)

***Attempt any 3 questions.**

Q21. Define cloud computing and list its essential characteristics.

Q22. What is the role of a hypervisor in cloud computing?

Q23. What is the significance of SLA in cloud computing?

Q24. Explain the concept of cloud bursting.

Q25. Why is cloud forensics more complex than traditional digital forensics?

Q26. Why is maintaining the chain of custody crucial in cloud forensics?

Long Answer Questions

(8 x 3 = 24)

***Attempt any 3 questions.**

Q27. Describe the NIST cloud computing reference architecture and its components.

Q28. Elaborate on the cloud forensic process and its phases.

Q29. An organization uses both AWS and Azure for different workloads. A security breach occurs, and forensic investigators are tasked with identifying the source and impact across both platforms. Discuss the challenges of conducting a forensic investigation in a multi-cloud environment. How would you correlate evidence across platforms, and what strategies would you use to ensure integrity and admissibility of evidence?

Q30. You are part of a forensic investigation team responding to a ransomware attack on a cloud-hosted application used by **XYZ Private Limited.** The organization operates in a **hybrid cloud environment**, utilizing both **AWS (IaaS)** and **Microsoft Azure (PaaS)**

for different workloads. The attack encrypted critical files and disrupted services. As part of the investigation, you have collected the following digital evidence:

VM Snapshot from AWS (2 GB)

Access Logs from Azure (500 MB)

Deleted file metadata from the cloud account

Cloud account information and login records

The evidence was acquired on **07 October 2025**, between **10:30 AM and 11:00 AM IST**, at the **ABC Forensic Lab, Bhopal, MP**. The tools used include **Magnet AXIOM** and **AWS CLI**. The evidence was analyzed on **08 October 2025**, and hash values were computed using **SHA-256**. The case is expected to proceed to court, and the evidence must be securely preserved and documented. **Prepare a detailed Chain of Custody (CoC) sheet** that includes the following sections:

Case Details

Description of Cloud Environment

Evidence Information (collection details, tools used, imaging status)

Evidence Description (analysis details, hash computation, storage)

Chain of Custody Transfers (who received and handed over the evidence)

Wrap-Up / Disposal (final disposition and secure erasure)

You may assume the following personnel were involved:

Mr. X, Lead Investigator

Miss Y, Forensic Analyst

Also explain the importance of maintaining a valid Chain of Custody in cloud forensic investigations and how it ensures the integrity and admissibility of digital evidence in legal proceedings.